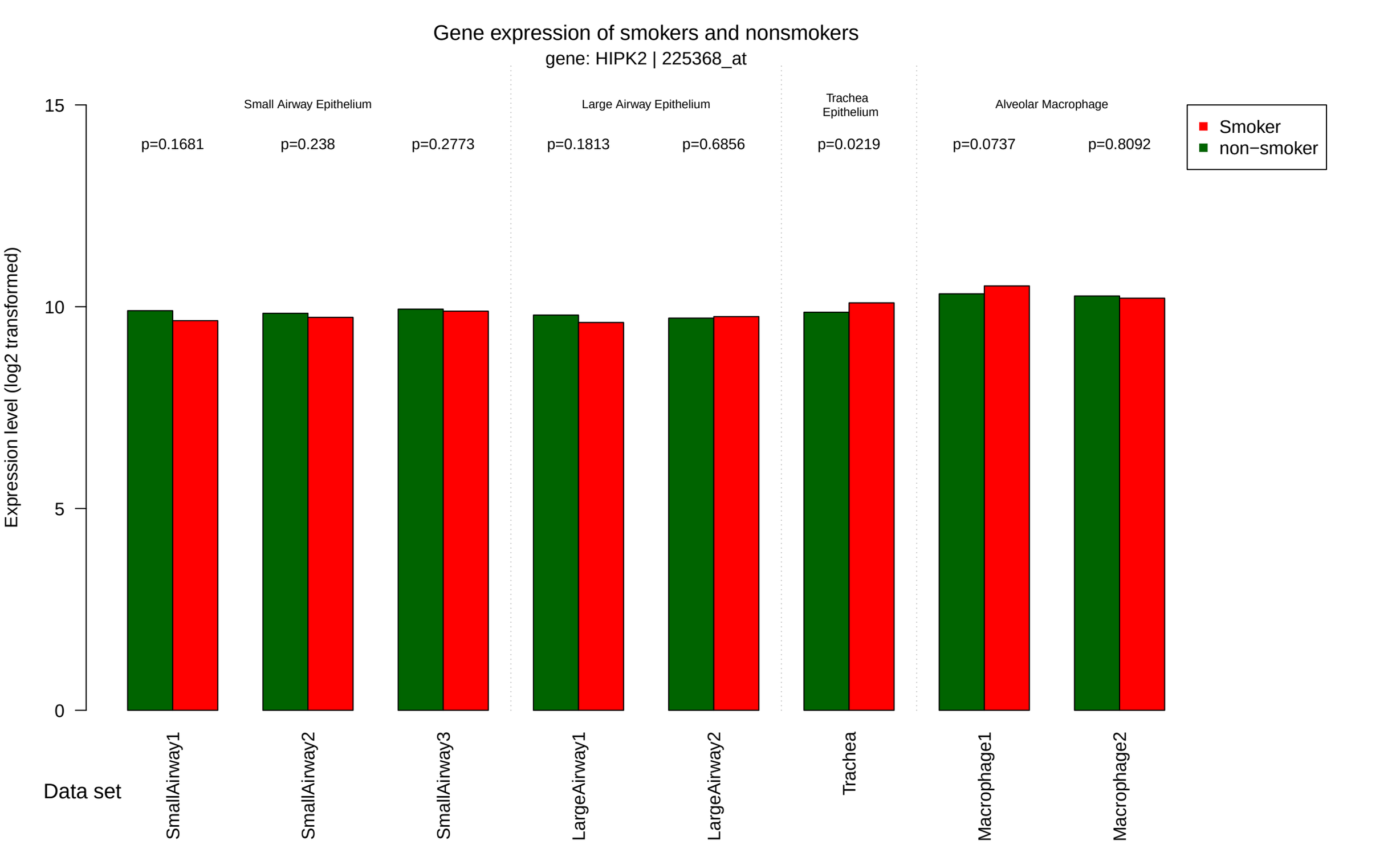
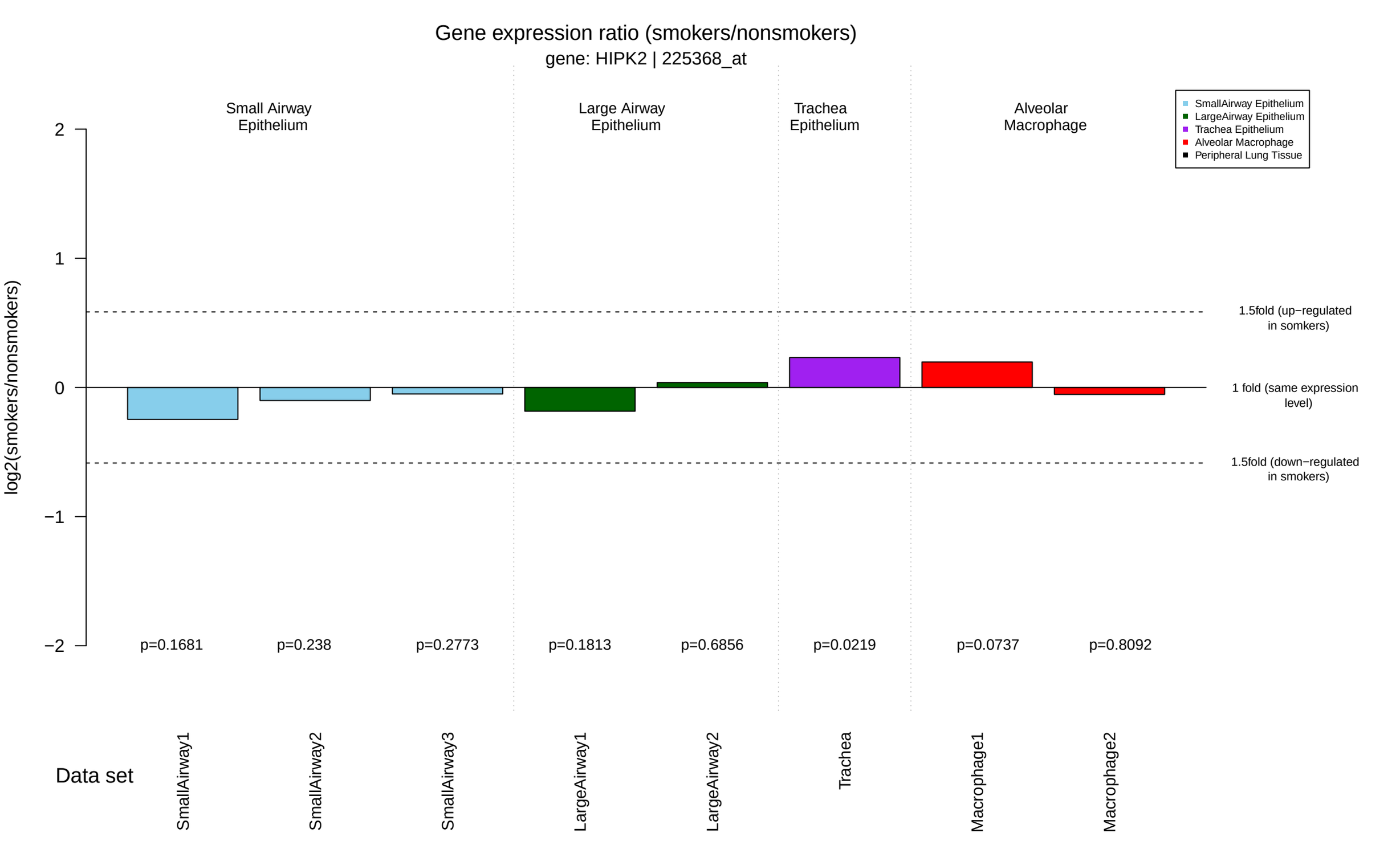
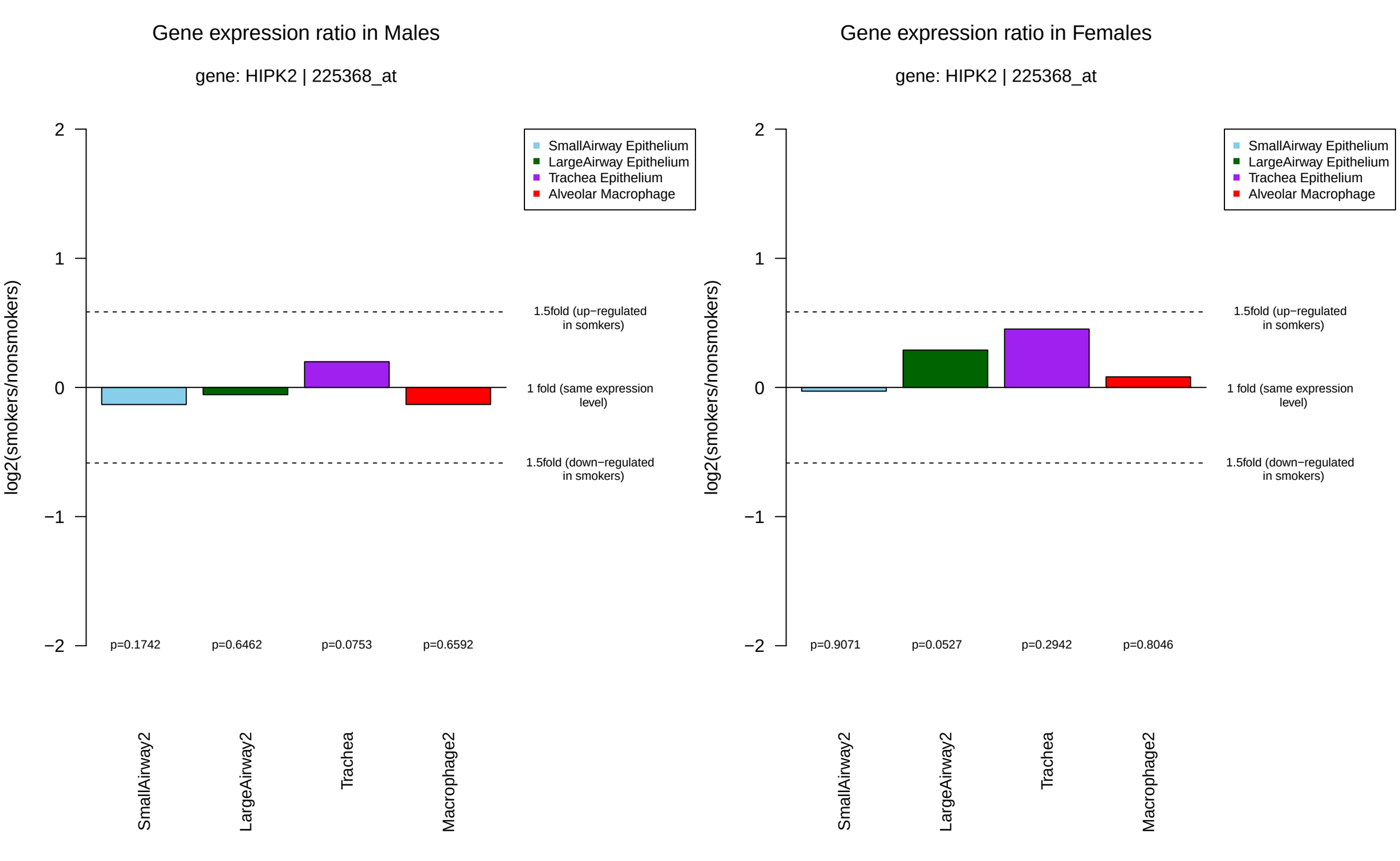
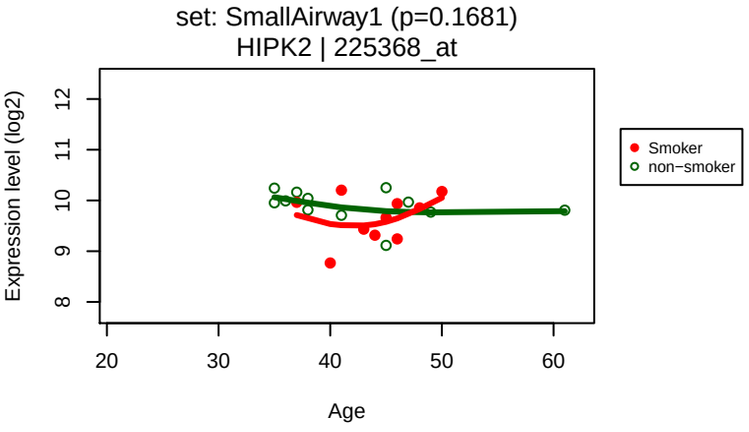
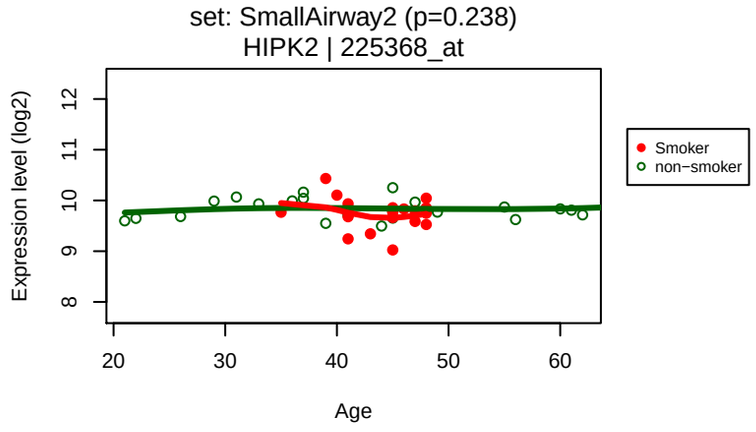
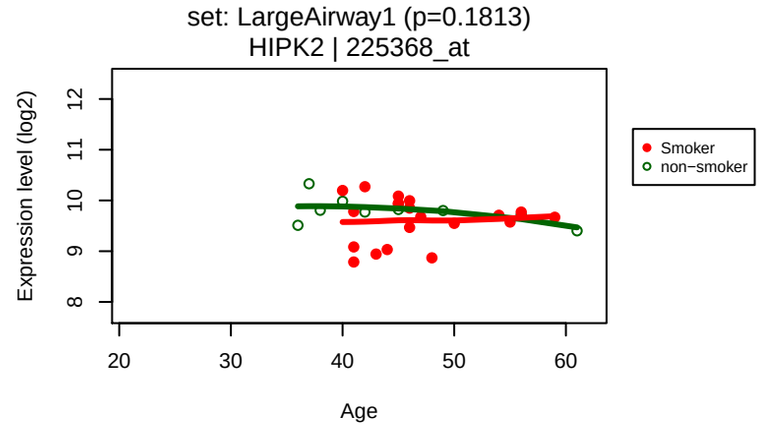

Gene expression of smokers and nonsmokers
gene: HIPK2 | 225368_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1681
p=0.238
p=0.2773
p=0.1813
p=0.6856
p=0.0219
p=0.0737
p=0.8092
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

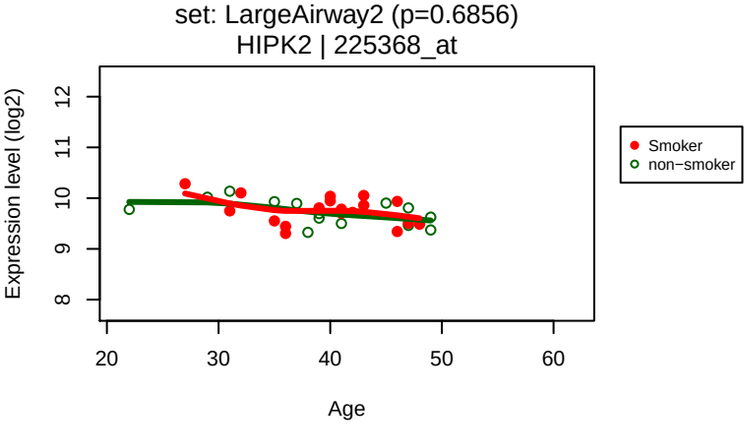

Gene expression ratio (smokers/nonsmokers)
gene: HIPK2 | 225368_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1681
p=0.238
p=0.2773
p=0.1813
p=0.6856
p=0.0219
p=0.0737
p=0.8092
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

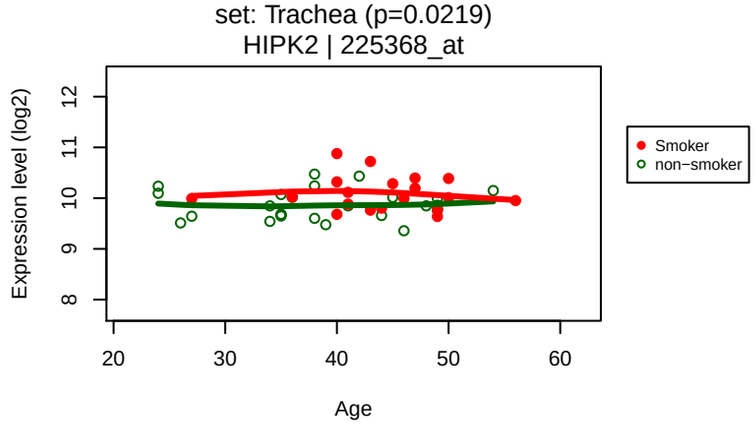

Gene expression ratio in Males
gene: HIPK2 | 225368_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1742
p=0.6462
p=0.0753
p=0.6592
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIPK2 | 225368_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9071
p=0.0527
p=0.2942
p=0.8046
SmallAirway2
LargeAirway2
Trachea
Macrophage2

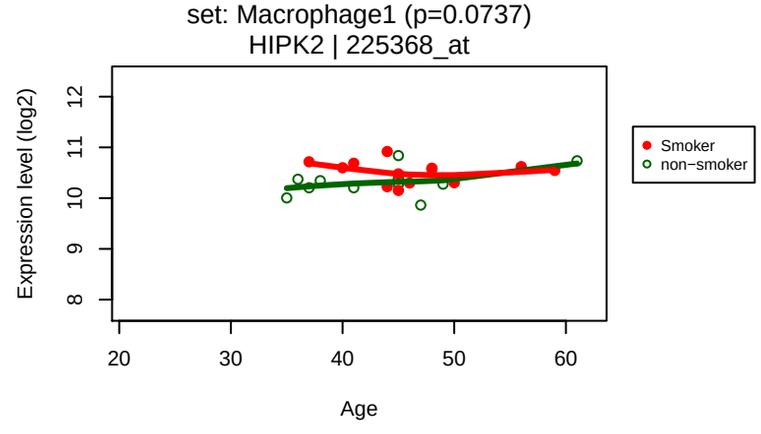

set: SmallAirway1 (p=0.1681)
HIPK2 | 225368_at
Expression level (log2)
Age
Smoker
non−smoker

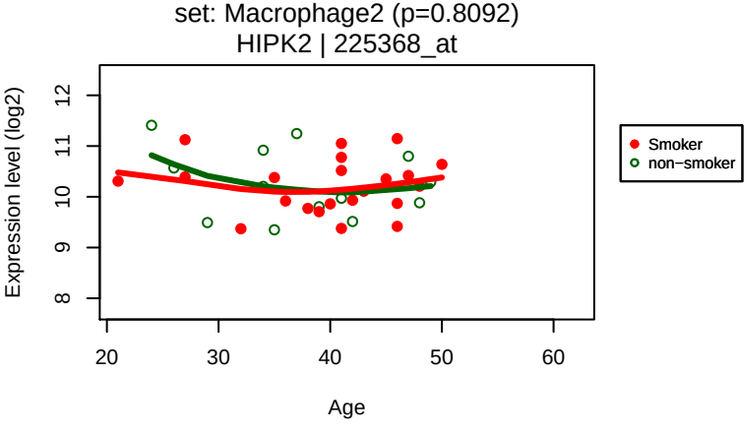

set: SmallAirway2 (p=0.238)
HIPK2 | 225368_at
Expression level (log2)
Age
Smoker
non−smoker

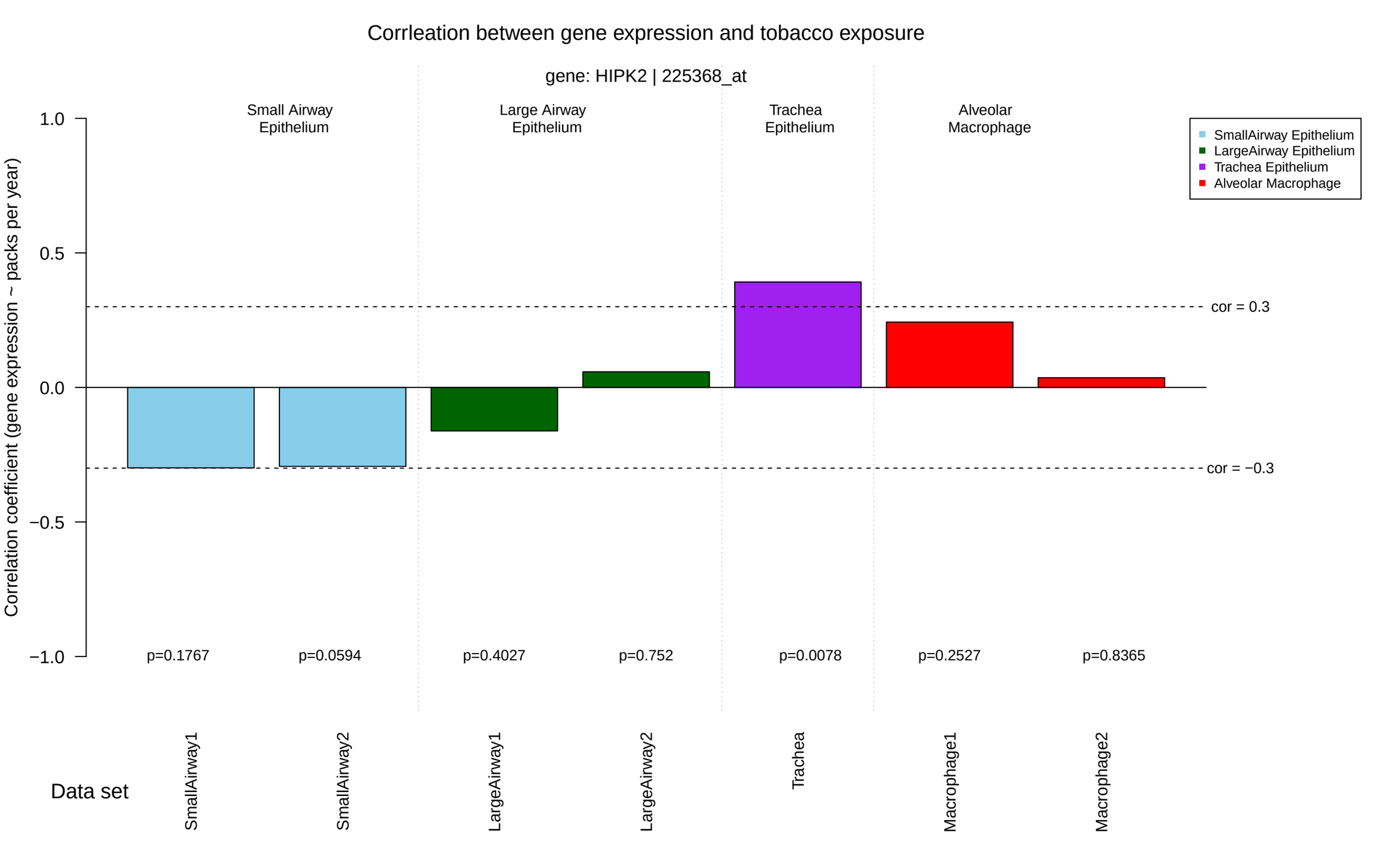

set: LargeAirway1 (p=0.1813)
HIPK2 | 225368_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6856)
HIPK2 | 225368_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0219)
HIPK2 | 225368_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0737)
HIPK2 | 225368_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8092)
HIPK2 | 225368_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIPK2 | 225368_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1767
p=0.0594
p=0.4027
p=0.752
p=0.0078
p=0.2527
p=0.8365
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2